

Experiment Report

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Course: Machine Learning

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Abstract

This study investigates the decision boundaries and model performance of various tree-based ensemble classifiers on the Iris dataset, including Decision Tree, Random Forest, Extra Trees, and AdaBoost classifiers. Each classifier was trained on different feature pairs to evaluate their decision boundaries and classification accuracy. Additionally, the study explores the impact of model depth and the number of estimators on optimizing tree-based models.

Introduction

Tree-based ensemble methods have gained widespread use in various machine learning applications due to their robustness, interpretability, and flexibility. The Iris dataset, a classic benchmark in pattern recognition, offers a suitable platform to explore and compare the decision surfaces of different tree-based classifiers. This study examines how different classifiers, including Decision Trees, Random Forests, Extra Trees, and AdaBoost, learn decision boundaries from the Iris dataset. By varying the feature pairs and model parameters, this work aims to provide a comprehensive comparison of these methods and their respective performances.

Materials and Methods

The dataset utilized in this study is the Iris dataset provided by scikit-learn. This dataset comprises 150 records, categorized into three distinct classes: Setosa, Versicolor, and Virginica. Each record features four attributes: sepal length, sepal width, petal length, and petal width. The study employs four models for classification: Decision Tree, Random Forest, Extra Trees, and AdaBoost.

In the experiments, each of the six classifiers is trained on different feature pairs (sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width), and their decision boundaries are plotted. Furthermore, the impact of varying *max_depth* (ranging from 1 to 100) and *n_estimators* (ranging from 1 to 100) parameters is examined to identify the configurations yielding the highest accuracy. The differences among the decision trees are analyzed and compared.

Results

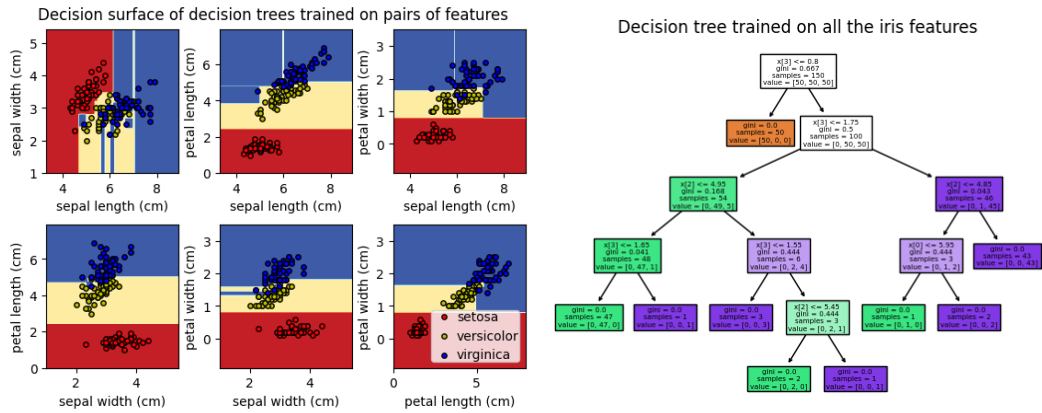


Figure 1. The left panel shows the decision boundaries plotted using the Decision Tree algorithm based on the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width. The right panel displays the corresponding decision trees.

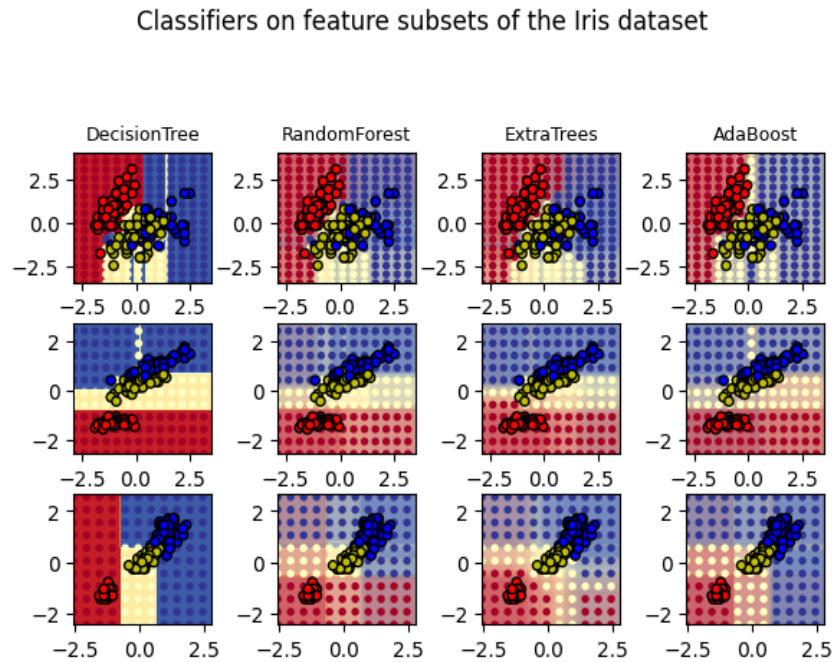


Figure 2. The chart depicts the decision boundaries obtained by applying the Decision Tree algorithm, Random Forest, Extra Trees, and AdaBoost classifiers to the data distributions based on sepal length vs. sepal width, sepal length vs. petal length, and petal length vs. petal width.

Classifiers on feature subsets of the Iris dataset

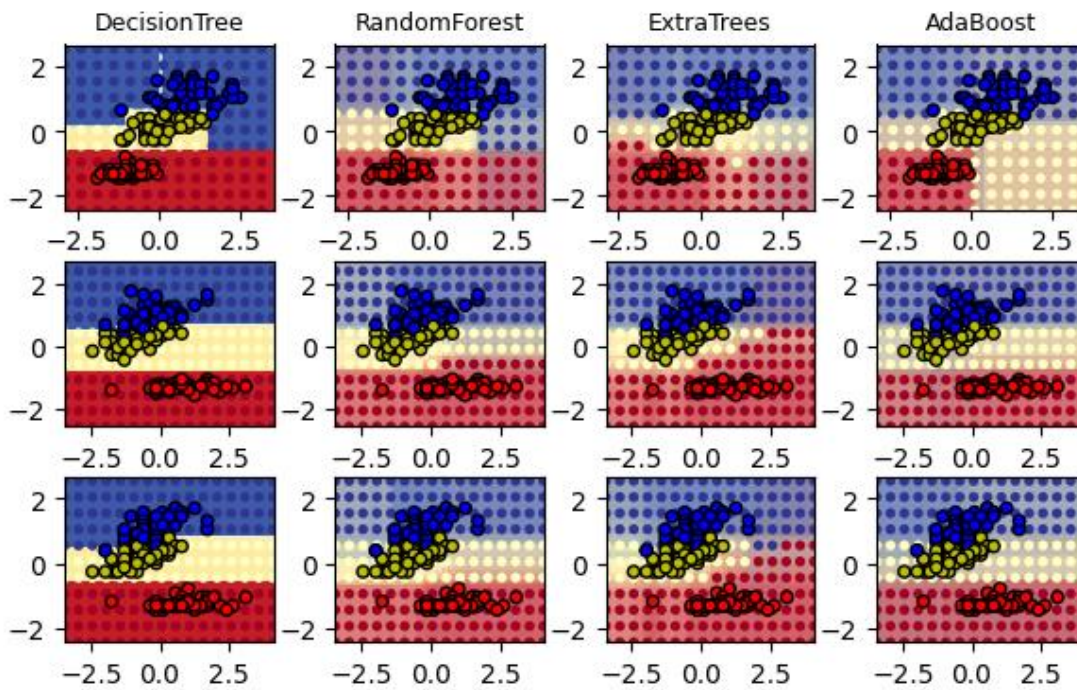


Figure 3. The chart illustrates the decision boundaries drawn using the Decision Tree algorithm, Random Forest, Extra Trees, and AdaBoost classifiers for the data distributions based on sepal length vs. petal width, sepal width vs. petal length, and sepal width vs. petal width.

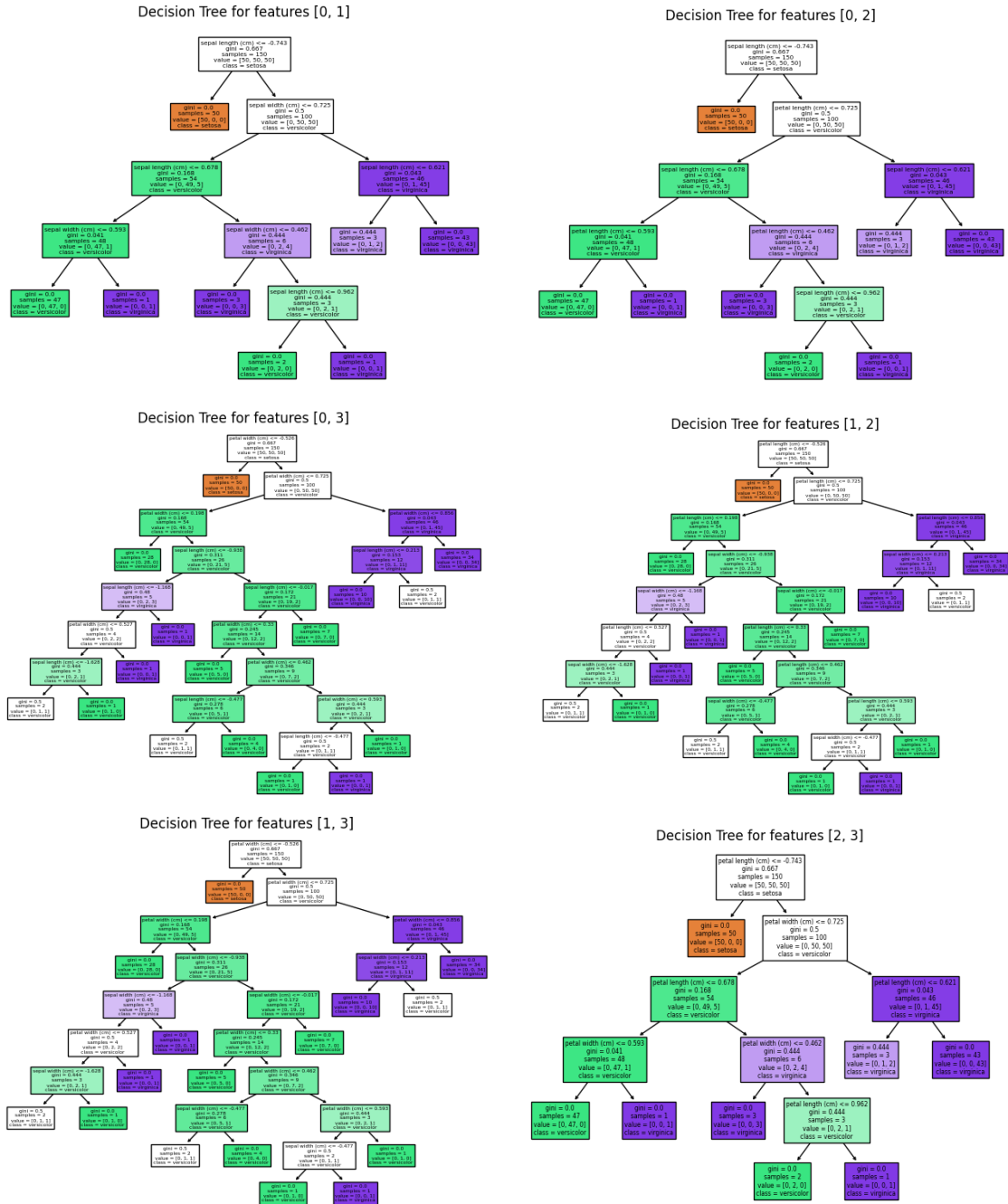


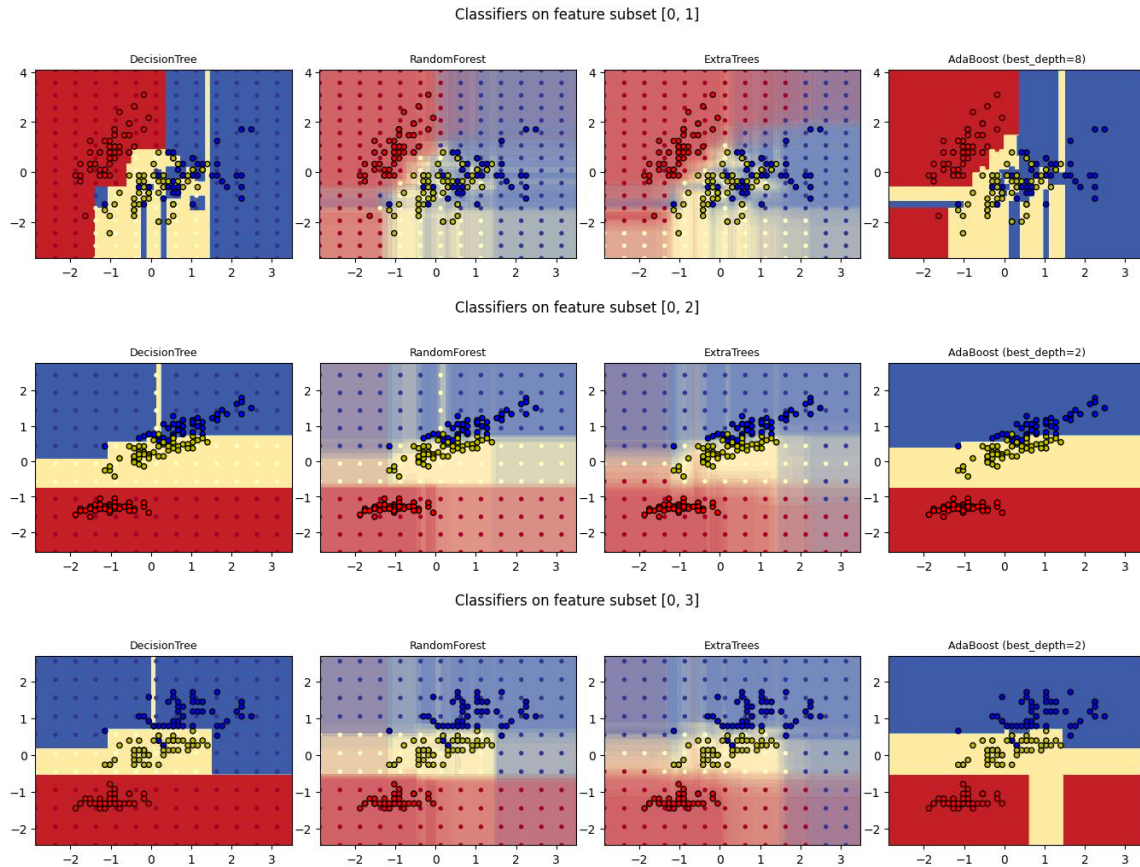
Figure 4. The chart shows the decision trees plotted using the Decision Tree algorithm for the data distributions based on sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width.

Table 1.
Accuracy comparison for various feature pairs across different algorithms.

Feature	DecisionTree	RandomForest	ExtraTrees	AdaBoost
[0, 1]	0.9267	0.9267	0.9267	0.8200
[0, 2]	0.9933	0.9933	0.9933	0.9933
[0, 3]	0.9733	0.9733	0.9733	0.9667
[1, 2]	0.9867	0.9867	0.9867	0.9867
[1, 3]	0.9800	0.9800	0.9800	0.9800
[2, 3]	0.9933	0.9933	0.9933	0.9867

Note.0=sepal length; 1: sepal width; 2: petal length;3: petal width

All figures are based on experiments conducted with $max_depth = 3$ and $n_estimators = 30$. Although the accuracy has reached nearly 100%, adjustments to max_depth will be made to identify models with potentially higher accuracy, while $n_estimators$ will remain fixed at 30.



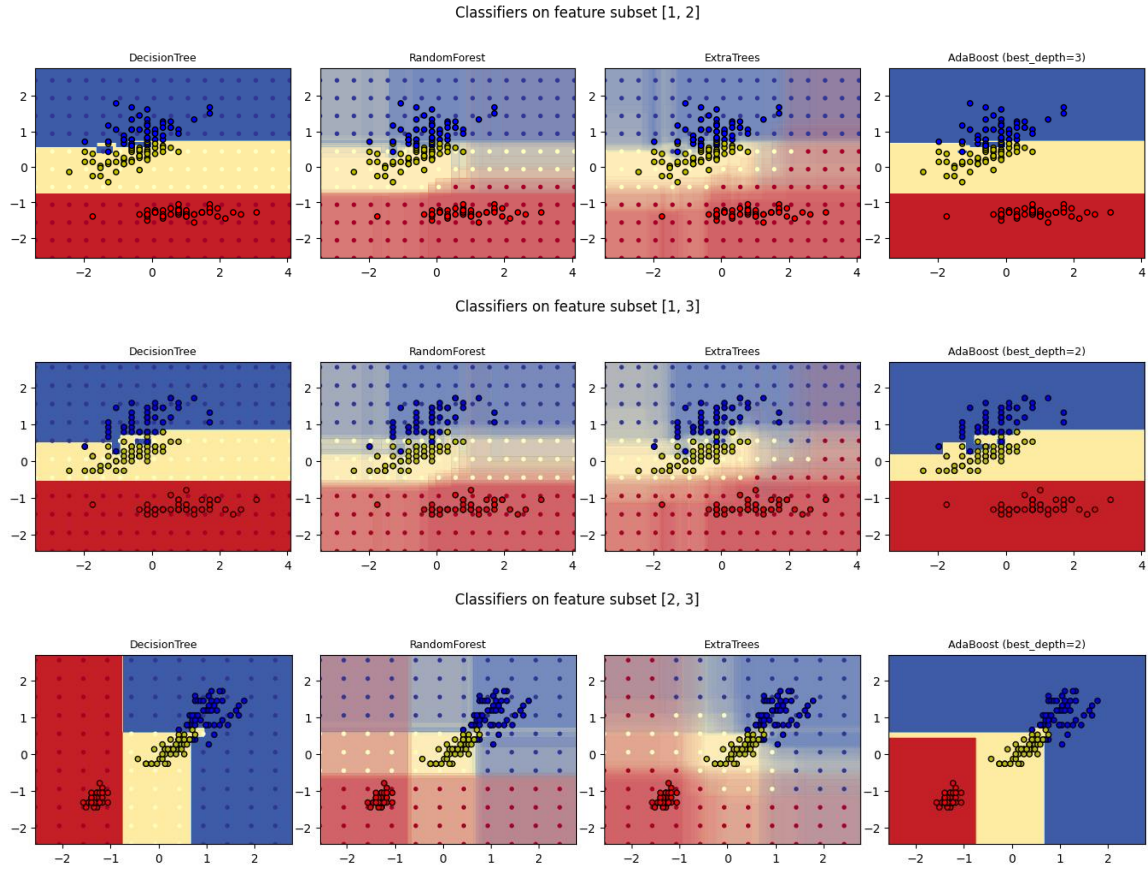
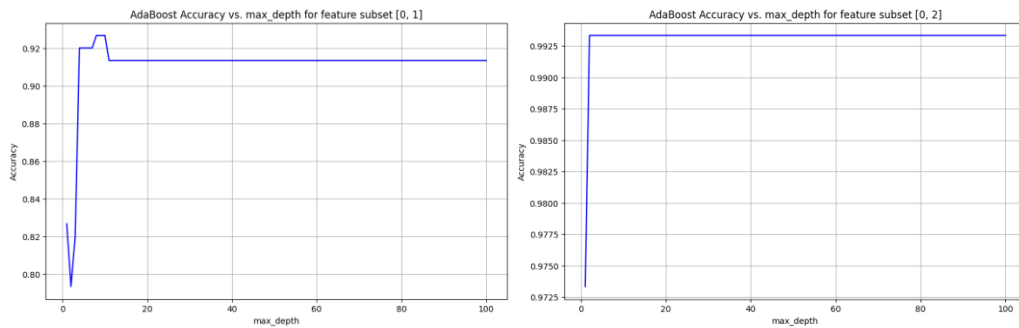


Figure 5. The upper panel shows the decision boundaries for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width, using Decision Tree, Random Forest, Extra Trees, and AdaBoost classifiers with *max_depth* adjusted to achieve the best accuracy.



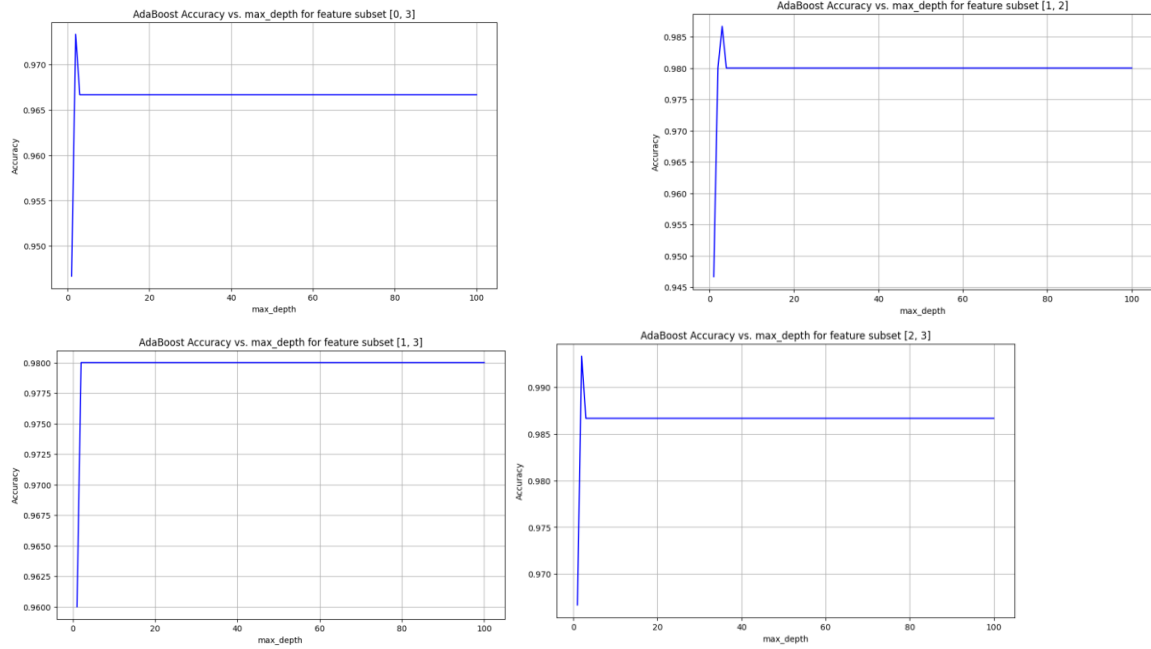


Figure 6. The upper panel illustrates the accuracy variation of AdaBoost with different values of *max_depth* (ranging from 1 to 100) for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width.

Table 2.

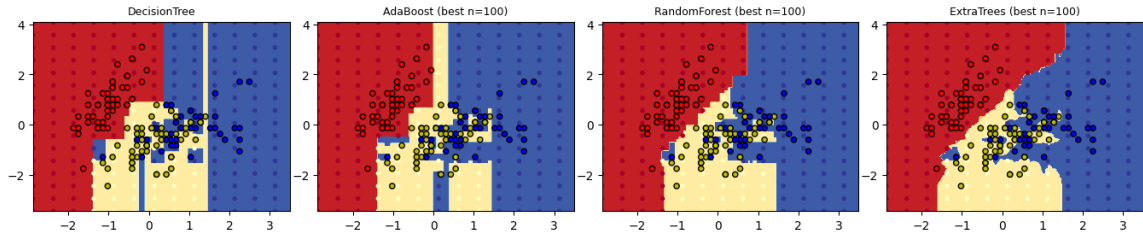
Accuracy comparison for various feature pairs across different algorithms.

Feature	DecisionTree	RandomForest	ExtraTrees	AdaBoost
[0, 1]	0.9267	0.9267	0.9267	0.9267
[0, 2]	0.9933	0.9933	0.9933	0.9933
[0, 3]	0.9733	0.9733	0.9733	0.9733
[1, 2]	0.9867	0.9867	0.9867	0.9867
[1, 3]	0.9800	0.9800	0.9800	0.9800
[2, 3]	0.9933	0.9933	0.9933	0.9933

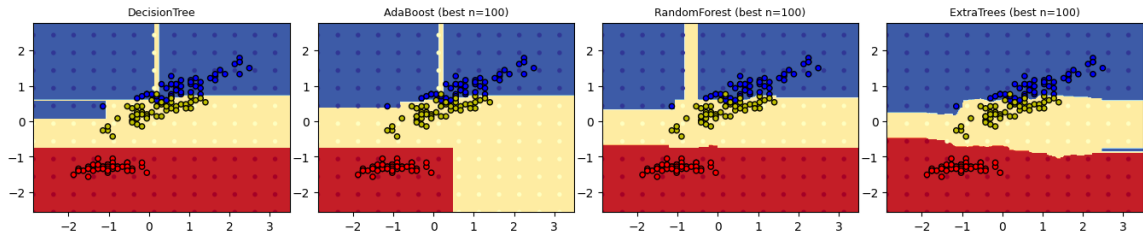
Note: 0 = sepal length; 1 = sepal width; 2 = petal length; 3 = petal width. The accuracy values for AdaBoost in the table are based on the results shown in Figure 5, with the highest accuracy selected.

All charts are based on experiments with *max_depth* ranging from 1 to 100. As seen in Table 2, although AdaBoost shows significant improvements in accuracy, adjustments to *n_estimators* will be made next, while *max_depth* will be fixed at 3 for these experiments.

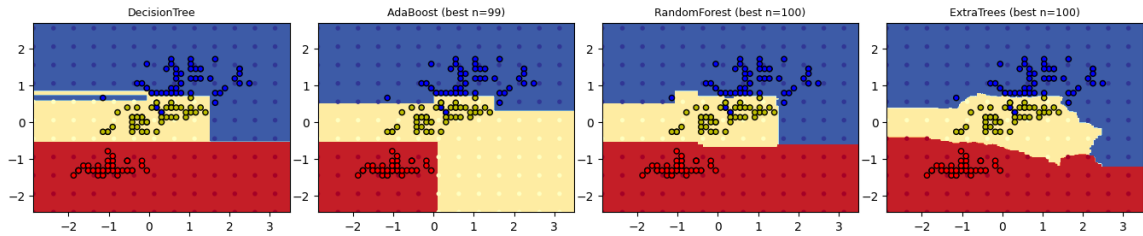
Classifiers on feature subset [0, 1]



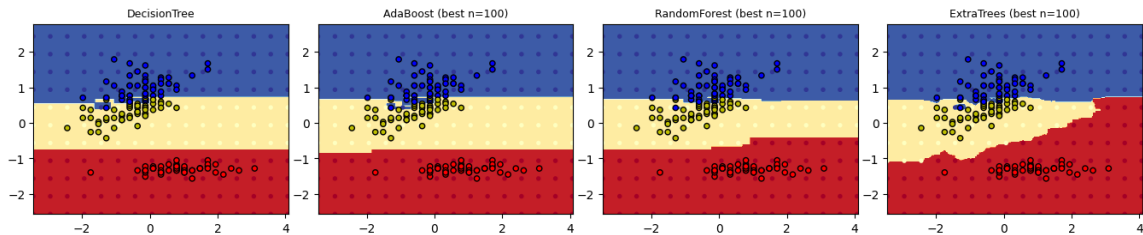
Classifiers on feature subset [0, 2]



Classifiers on feature subset [0, 3]



Classifiers on feature subset [1, 2]



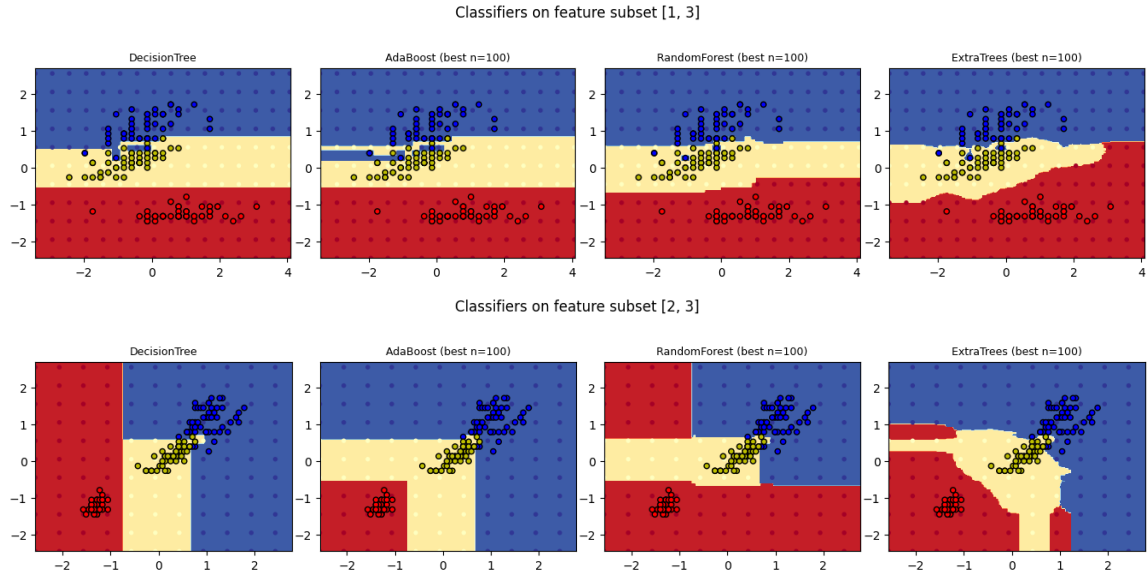
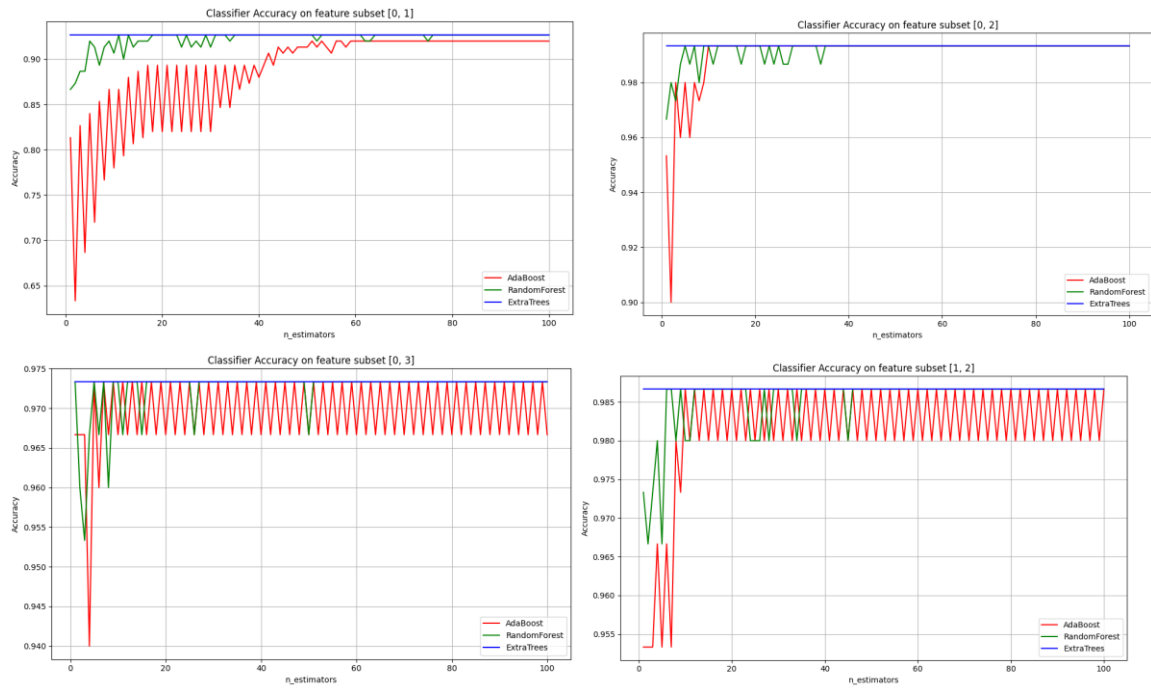


Figure 7. The upper panel displays the decision boundaries for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width, using Decision Tree, Random Forest, Extra Trees, and AdaBoost classifiers with $n_estimators$ adjusted to achieve the highest accuracy.



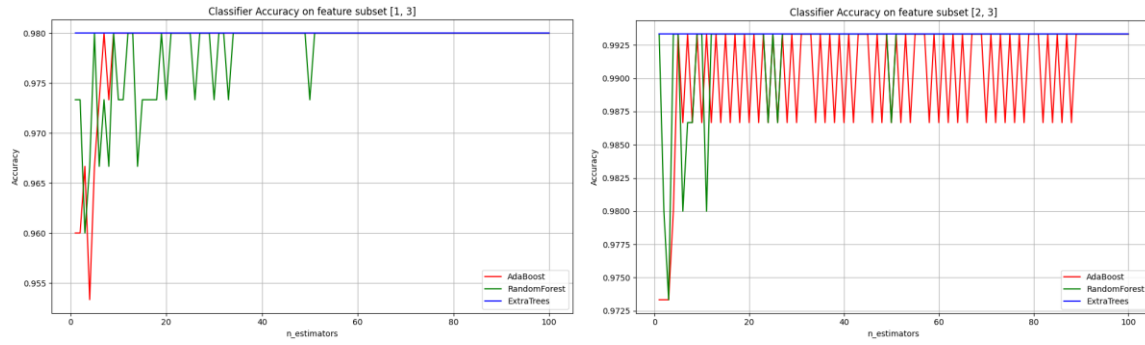


Figure 8. The upper panel illustrates the accuracy variations of Random Forest, Extra Trees, and AdaBoost with different values of $n_estimators$ (ranging from 1 to 100) for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width.

Table3

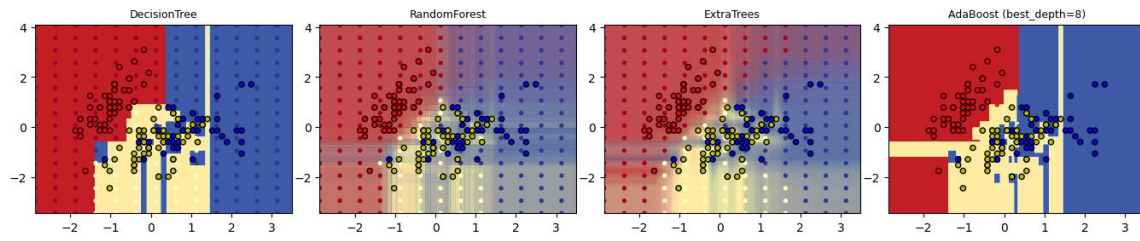
Accuracy comparison for various feature pairs across different algorithms.

Feature	DecisionTree	RandomForest	ExtraTrees	AdaBoost
[0, 1]	0.9267	0.9267	0.9267	0.9200
[0, 2]	0.9933	0.9933	0.9933	0.9933
[0, 3]	0.9733	0.9733	0.9733	0.9733
[1, 2]	0.9867	0.9867	0.9867	0.9867
[1, 3]	0.9800	0.9800	0.9800	0.9800
[2, 3]	0.9933	0.9933	0.9933	0.9933

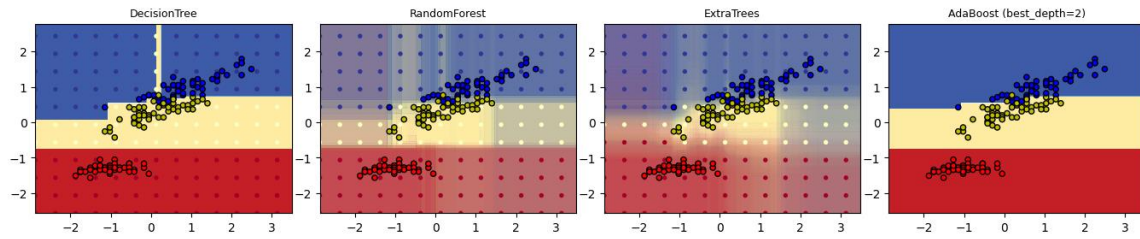
Note. 0 = sepal length; 1 = sepal width; 2 = petal length; 3 = petal width. The accuracy values for Random Forest, Extra Trees, and AdaBoost in the table are based on the results shown in Figure 8, with the highest accuracy selected.

All figures are based on experiments with $n_estimators$ ranging from 1 to 100. As seen in Table 3, although the accuracy did not show a significant improvement compared to Table 2, adjustments to $n_estimators$ will be explored further. Next, $n_estimators$ will be set to 100, and max_depth will be adjusted.

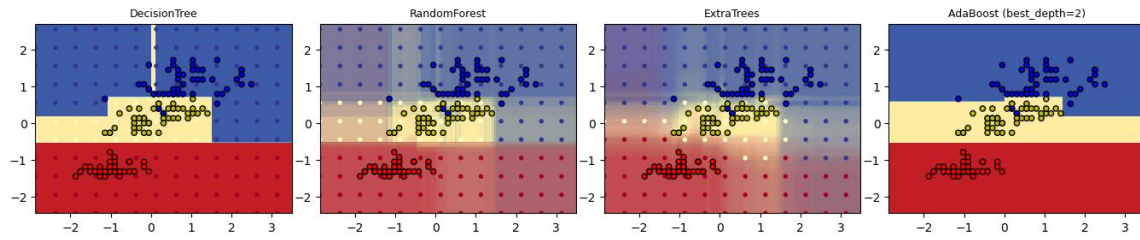
Classifiers on feature subset [0, 1]



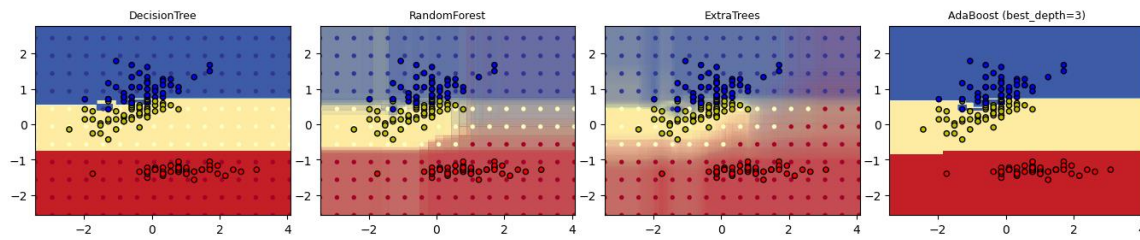
Classifiers on feature subset [0, 2]



Classifiers on feature subset [0, 3]



Classifiers on feature subset [1, 2]



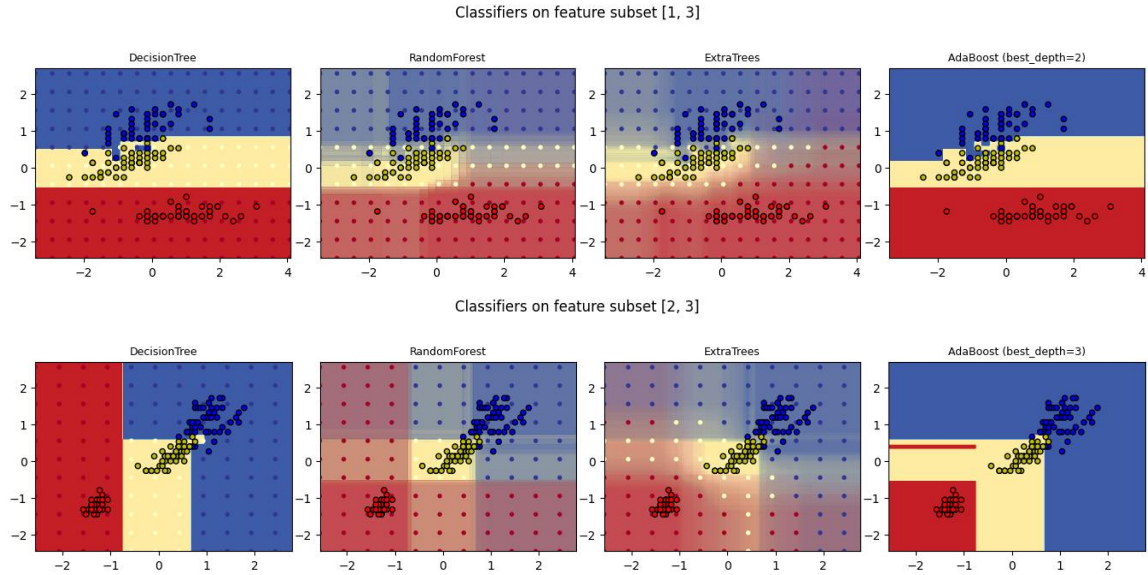
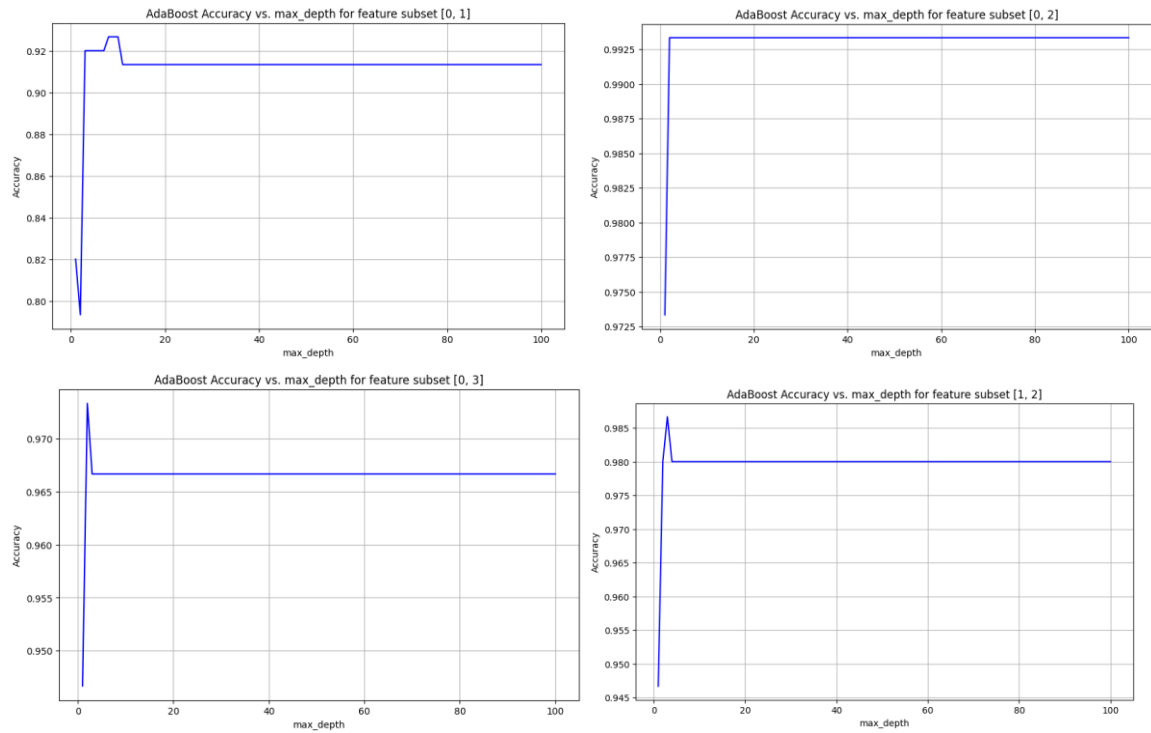


Figure 9. The upper panel displays the decision boundaries for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width, using Decision Tree, Random Forest, Extra Trees, and AdaBoost classifiers with *max_depth* adjusted to achieve the highest accuracy.



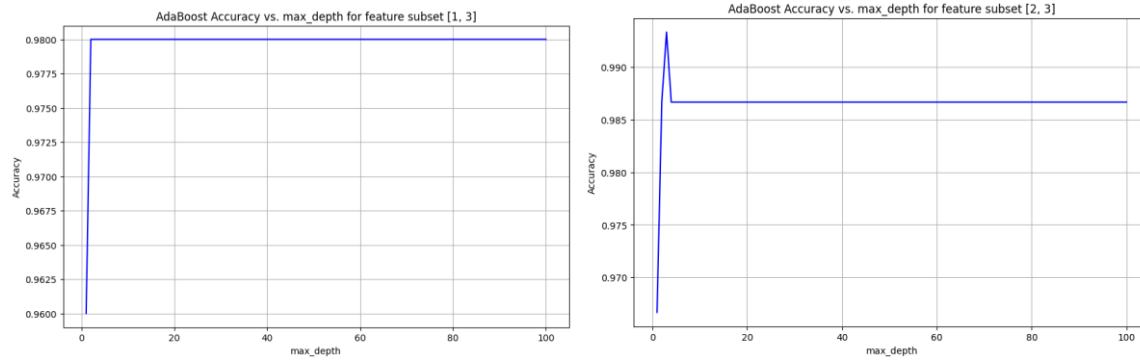


Figure 10. The upper panel illustrates the accuracy variations of AdaBoost with different values of *max_depth* (ranging from 1 to 100) for the feature pairs: sepal length vs. sepal width, sepal length vs. petal length, sepal length vs. petal width, sepal width vs. petal length, sepal width vs. petal width, and petal length vs. petal width.

Table4

Accuracy comparison for various feature pairs across different algorithms.

Feature	DecisionTree	RandomForest	ExtraTrees	AdaBoost
[0, 1]	0.9267	0.9267	0.9267	0.9267
[0, 2]	0.9933	0.9933	0.9933	0.9933
[0, 3]	0.9733	0.9733	0.9733	0.9733
[1, 2]	0.9867	0.9867	0.9867	0.9867
[1, 3]	0.9800	0.9800	0.9800	0.9800
[2, 3]	0.9933	0.9933	0.9933	0.9933

Note. 0 = sepal length; 1 = sepal width; 2 = petal length; 3 = petal width. The accuracy values for AdaBoost in the table are based on the results shown in Figure 10, with the highest accuracy selected.

Discussion

The results indicate that when feature combinations are highly correlated (e.g., sepal length vs. petal length), adjustments to either *n_estimators* or *max_depth* show minimal differences in model performance. However, when using less correlated feature pairs (e.g., sepal width vs. sepal length), moderate adjustments to *n_estimators* or *max_depth* can slightly improve accuracy, though they eventually reach a plateau, suggesting that the model's accuracy may be near its maximum limit.

Conclusion

This study compared the classification performance of six models on the Iris dataset. Random Forest and Extra Trees demonstrated more stable and accurate performance, especially achieving near-perfect classification results with sepal length and petal length features. AdaBoost showed some performance degradation with more independent sample data but still holds potential. Future work could focus on increasing the sample size and conducting more in-depth parameter optimization for each model.

References

1. Scikit-learn. (n.d.). Random forests on the Iris dataset. Retrieved from https://scikit-learn.org/stable/auto_examples/ensemble/plot_forest_iris.html
2. Scikit-learn. (n.d.). Decision tree classification on the Iris dataset. Retrieved from https://scikit-learn.org/stable/auto_examples/tree/plot_iris_dtc.html